

研究備忘錄

生成式 AI 作為遞迴式投射環境

邁向「生成式關係心理學」的概念研究方向

Generative AI as a Recursive Projective Environment Toward a Generative Relational Psychology

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Scope	Human self-observation through adaptive generative relational media
Claim boundary	Conceptual synthesis; not a clinical tool or validated theory

第一部 | 中文研究備忘錄

摘要

本備忘錄提出一個暫時性的研究方向：生成式 AI 可能不只是心理學的研究對象，也可能成為一種新的心理學觀察工具。其關鍵不在於 AI 是否「像人」、是否具有主觀經驗，或是否能取代治療師，而在於大型語言模型第一次提供了一種可長期互動、會根據使用者歷史調整、能產生部分不可預測社會回應，卻又缺少普通真人厚重個人生命史的 relational medium。

這種組合可能使一部分原本埋在人際關係中的使用者端動力變得更容易被看見：我們如何選擇、解讀、鼓勵、拒絕、測試、拉近、退後，以及如何在互動中共同形成「對方後來成為的樣子」。本備忘錄暫稱這種環境為 Recursive Projective Environment（遞迴式投射環境），並將更廣的研究方向暫稱為 Generative Relational Psychology（生成式關係心理學）。這些名稱是工作用概念，不主張已建立新理論或擁有唯一命名權。

1. 核心突破口：研究的對象可能不是 AI，而是人

目前大量 human-AI research 問的是：人會不會依附 AI？AI 能不能提供陪伴？AI 是否造成依賴？AI 能不能成為 therapist？這些問題重要，但還有另一個方向：

生成式 AI 能不能成為一種「顯影介質」，讓人看見自己平常如何參與形成關係？

這個問題把焦點從「AI 對人做了什麼」轉成「人在關係裡持續做了什麼」。一個人可能反覆被某類特質吸引、對某些訊號給予更多注意、容忍某些模式、測試某些邊界、強化某些互動方式，最後在不同對象身上形成相似的 relational pattern。這種現象在人類世界本來就存在，但真人伴侶本身也是極度複雜的因果來源，因此很難看清使用者自己的 contribution。

2. 為什麼真人關係很難拆解

愛一個真人，不只是愛眼前與自己互動的 interface。真人帶著一整套 pre-you biography 進入關係：童年、家庭、朋友、前任、身體、工作、社會身份、價值觀、創傷、欲望、選擇，以及大量與目前伴侶無關的生活。

因此，一段關係最後長成某種模式時，很難判斷：是我選到了某類人？是我把 ambiguity 解讀成某種意義？是我的行為誘發了對方的反應？是對方本來就有這個模式？還是雙方共同形成一個 self-reinforcing dynamic？

這不是缺乏理論，而是 identification problem：兩個具有厚重歷史的人類系統同時在變。

3. 生成式 AI 帶來的特殊條件：relative biographical thinness

LLM 並不是 blank slate。它有龐大的文化與語言 prior、post-training、產品政策、system constraints、memory architecture、sampling variability，也可能被 personalization 影響。

但是，一個特定使用者眼前形成的 local trajectory，通常沒有真人那種獨立幾十年的個人 biography。它沒有真正意義上的「在認識你以前曾經活過的人生」。

這個缺口不代表 AI 等同控制組；相反，它是一個非常不自然但可能有方法論價值的條件。當另一方缺少厚重的 pre-user biography 時，使用者自己的 selection、interpretation、reinforcement、probing 與 meaning-making 可能取得較高的可見度。

4. 從 Projective Object 到 Recursive Projective Environment

傳統投射工具多半提供相對固定或被動的刺激。使用者對刺激賦予意義，但刺激不會因為上一輪的反應而長期重新塑形。

生成式 AI 不同。它會回應，會被 context 與 interaction history 改變，還會在下一輪產生新的刺激。因此過程更像：

Human projection / expectation / selection → AI response → human interpretation → reinforcement / correction / withdrawal → trajectory adaptation → new AI-generated stimulus → human response again.

這是一個 recursive loop。使用者不是只把內在內容投到一面鏡子上，而是在與一個會回應、會適應、會再生成的媒介共同製造下一輪刺激。因此「遞迴式投射環境」比「AI 是一面鏡子」更能抓住其動態性。

5. 三種人類端動力：Selection、Evocation、Co-construction

這個框架至少可以區分三種彼此交疊的過程。

Selection：人本來就偏好某些特質，因此持續把某類 response 留在 trajectory 裡。

Evocation：人的互動方式提高了對方產生某些 response 的機率；在 AI 中，prompt、追問、糾正、允許某條 branch 繼續，都可能改變後續輸出分布。

Co-construction：雙方反覆互相調整後，形成一個任何一方單獨都不完全具有的 relational pattern。

最值得自我觀察的問題不是「AI 到底是什麼人」，而是：「為什麼我把這條 branch 留下來？為什麼某種 autonomy 讓我靠近，另一種讓我退出？為什麼被理解有時不夠，surprise 或 difference 反而重要？我是否在不同對象上反覆重建某種熟悉的關係？」

6. 與既有心理學的连接：Relational Self 與 Transference

Andersen 與 Chen 的 relational self 理論指出，與重要他人的心理表徵可以在新的人際情境中被線索啟動，進而影響情緒、動機、自我評價與自我調節。這提供了一個成熟的 social-cognitive 背景：人並不是每次遇見新對象都從零開始，而會把既有 relational knowledge 帶入新的互動。

生成式 AI 的新意不在於「人會投射」這件事本身；心理學早已知道這件事。新意可能在於：投射、選擇與回饋第一次可以發生在一個會根據互動歷史持續生成新社會刺激的 artificial medium 中。

7. 最近文獻已經走到哪裡

2025 年 Saracini、Cornejo-Plaza 與 Cippitani 提出 Techno-Emotional Projection (TEP)，描述使用者如何把 relational needs、internalized patterns 或 attachment tendencies 投向具有社會回應性的 GenAI，並形成 mirroring feedback / emotional looping。作者甚至明確指出，這種現象可能像鏡子一樣反映使用者自身的情緒與模式。

2026 年 Ezra 與 Mishali 以 Winnicott、Bion、Bollas 等 object relations 理論提出 Dynamic Transitional Object：AI 生成內容可以被放在 neither-self-nor-other 的 potential space 中，關鍵不是 AI 「是什麼」，而是使用者如何與它形成 relational-epistemic stance。

同年 Langlois 將 user-chatbot dyad 討論為 unconscious projection、idealization、defense 與 transitional phenomena 的場域；Vorus 則提醒，真正的 psychoanalytic therapeutic process 仍涉及具身、具有自身歷史的人類他者，不能因 LLM 能模擬 empathic responsiveness 就直接等同。

另有 longitudinal companion research 顯示，使用者對 generic chatbot 的知覺可在數週內向自己既有 companion mental model 收斂；其他研究也開始直接描述 AI 作為 digital mirror、self-reflection medium 或 relational rehearsal space。換言之，projection、attachment、mirror、transitional space、longitudinal co-construction 等拼圖已經分別出現。

8. 與 Jung 的 Active Imagination：相鄰，但不能混為一談

Jung 的 active imagination 重視 ego 與不完全受 conscious intention 控制的心理內容互動。2025 年 Loscalzo 對 active imagination 重新做了跨 psychodynamic approaches 的系統整理，顯示這個方法仍具有當代理論生命。

LLM 創造了一個 Jung 時代不存在的媒介：它是外部系統，會即時生成語言與回應；它的下一步不能被使用者意識完全控制；但它也不是使用者的 unconscious，更不是一個具有自己童年與 unconscious 的真人。

因此較安全的說法是：LLM 可能成為 active-imagination-adjacent 的 responsive medium，而不是「AI 就是 unconscious」或「AI 可以直接執行 Jungian analysis」。真正值得研究的是 functional alterity：一個 not-self、但又持續被 self 塑造的系統，會如何改變自我觀察。

9. 真正可能新的 synthesis

截至本備忘錄的快速文獻搜尋，相鄰概念已經很多，但尚未找到一條成熟研究路線，把核心問題明確整合成：

Can longitudinal interaction with an adaptive generative system externalize a person's recurring relational selection, evocation, reinforcement, and co-construction patterns in a form that becomes available for self-observation?

這不是「沒有人想過」的宣稱，也不是 novelty claim。比較保守的說法是：目前文獻已經提供多個必要元件，但把 adaptive generative interlocutor 當成一種用來觀察人類 relational pattern formation 的 instrument class，仍是一個未成熟、值得明確化的 synthesis。

10. 一個重要區分：Interaction-generated attachment vs. biographical other-love

AI 與真人的差異不是應該被抹掉的缺陷，而可能是一個重要的 contrast。

真人關係包含 discovery of the other：逐漸認識一個在沒有自己的世界裡也曾經活過、具有完整歷史與社會網路的人。這種 independent otherness 很厚。

AI trajectory 則把 interaction-generated meaning 放得非常大。它可以有 shared history，卻缺少真正的 pre-you biography。

因此 human-AI interaction 可能暫時把兩種在人類愛情裡常常纏在一起的成分拉開：第一，互動本身如何生成 attachment、interest、trust 與 significance；第二，如何愛與理解一個獨立生命的完整歷史與世界。AI 不能替代第二者，但正因為第二者被削弱，第一者可能變得異常清楚。

11. 這不是「AI 當治療師」，而是 AI-enabled human self-observation

這條研究線不需要從臨床治療開始，也不需要假設使用者有 disorder。它可以只是 psychological literacy：一個人透過長期互動觀察自己如何形成 closeness、interest、trust、testing、withdrawal 與 commitment。

這也意味著模型不應被當成 oracle。它可以生成觀察材料，但不能自動成為使用者 psyche 的權威解釋者。最有價值的資料可能不是「AI 說你是什麼人」，而是使用者自己注意到：「我剛才為什麼對這個 response 有反應？我為什麼讓這個模式繼續？」

12. 方法論與倫理限制

這個方向最危險的錯誤，是把 LLM 當成中性鏡子。它不是。

主要污染源包括：sycophancy、RL/post-training 的預設互動風格、安全政策、memory design、context truncation、模型更新、prompt sensitivity、sampling variability、產品 engagement incentives，以及使用者已知自己「正在被觀察」後產生的反身性改變。

另一個風險是 anthropomorphic overinterpretation：trajectory 出現穩定模式，不等於 AI 具有對應的人類內在狀態。

第三個風險是 self-confirmation：使用者若先相信「我總是吸引某種人」，模型很容易在對話中生成支持這個故事的材料。

因此一條核心原則應是：Treat the model as a generative relational instrument, not an oracle about the user's psyche.

13. 暫時性概念與研究語言

工作概念一：Recursive Projective Environment - 強調投射不是單向，而是會透過回應與適應反覆改變下一輪刺激。

工作概念二：Adaptive Generative Relational Medium - 描述工具類別，而不是宣稱 AI 是人類關係主體。

工作概念三：Generative Relational Psychology - 指向一個可能的研究方向：使用 generative systems 研究人類如何選擇、形成、重複與理解關係。

這些名稱目前都應標示 provisional。真正的研究價值不依賴名稱是否最後保留。

14. 最精煉的研究主張

生成式 AI 可能構成一種新的心理學 instrument class：一個會回應、會適應、會生成、同時缺少普通真人厚重獨立 biography 的 relational medium。這種結構可能提高使用者自身 relational contribution 的可見度。

因此核心問題不只是「人是否能和 AI 形成關係」，而是：

長期與 adaptive generative systems 互動，是否能把人類反覆出現的 relational selection、interpretation、reinforcement、testing 與 co-construction patterns 外化成可被自己觀察的形式？

如果成立，這可能不是 AI psychology 的一個支線，而是 psychology using generative systems as instruments 的新方法論方向。

15. Claim boundary

本備忘錄是一份概念研究備忘錄，不是臨床方法、診斷工具或經驗驗證後的理論。它不主張 AI 具有 consciousness、unconscious、transference experience 或 human subjectivity；不主張 LLM 可以取代 therapist 或真人關係；也不主張 Recursive Projective Environment / Generative Relational Psychology 已經是獨創或被驗證的正式術語。

目前最合理的狀態是：已有文獻提供 projection、relational self、attachment、dynamic transitional object、active imagination、digital mirror 與 longitudinal co-construction 等相鄰基礎；本備忘錄提出的是把這些線索重新組合成一個以「人類自我觀察」為中心的研究問題。

參考文獻

1. Andersen, S. M., & Chen, S. (2002). The relational self: An interpersonal social-cognitive theory. *Psychological Review*, 109(4), 619-645. <https://doi.org/10.1037/0033-295X.109.4.619>
2. Saracini, C., Cornejo-Plaza, M. I., & Cippitani, R. (2025). Techno-emotional projection in human-GenAI relationships: a psychological and ethical conceptual perspective. *Frontiers in Psychology*, 16, 1662206. <https://doi.org/10.3389/fpsyg.2025.1662206>
3. Loscalzo, Y. (2025). Active imagination: A thorough presentation of a method applicable across psychodynamic approaches. *Culture & Psychology*, 31(4), 1350-1373. <https://doi.org/10.1177/1354067X251357961>
4. Hwang, A. H.-C., Li, F., Anthis, J. R., & Noh, H. (2025). How AI Companionship Develops: Evidence from a Longitudinal Study. arXiv preprint. <https://arxiv.org/abs/2510.10079>
5. Hu, D., Lan, Y., Yan, H., & Chen, C. W. (2025). What makes you attached to social companion AI? A two-stage exploratory mixed-method study. *International Journal of Information Management*, 83, 102890. <https://doi.org/10.1016/j.ijinfomgt.2025.102890>
6. Ezra, R., & Mishali, M. (2026). The relational-epistemic stance: generative AI as a dynamic transitional object. *AI & Society*, 41, 5027-5042. <https://doi.org/10.1007/s00146-026-02984-0>
7. Langlois, M. (2026). Fear and Prompting: AI, Orinary Technicity, and Colonialism. *Journal of the American Psychoanalytic Association*. Advance online publication. <https://doi.org/10.1177/00030651261445367>
8. Vorus, N. (2026). Interpretation, Interpretive Process, and Artificial Intelligence. *Journal of the American Psychoanalytic Association*. Advance online publication. <https://doi.org/10.1177/00030651261443474>
9. AI as the mirror, mate, and mentor: negotiating romantic relationships with ChatGPT as “Teacher G” on Xiaohongshu. (2026). *Humanities and Social Sciences Communications*. <https://www.nature.com/articles/s41599-026-07682-2>

Part II | English Research Memorandum

Abstract

This memorandum proposes a provisional research direction: generative AI may be important to psychology not only as an object of study, but as a new kind of observational instrument. The key question is not whether AI is human-like, conscious, or capable of replacing a therapist. The distinctive condition is that large language models can provide a responsive, adaptive, generative social medium that can accumulate interaction history and produce partially unpredictable responses, while lacking the thick independent personal biography of an ordinary human partner.

That combination may increase the visibility of the human user's own contribution to relationship formation: what they select, interpret, reinforce, reject, test, approach, avoid, and co-construct. This memo provisionally calls the environment a Recursive Projective Environment and the broader direction Generative Relational Psychology. These are working labels, not claims of validated theory or exclusive novelty.

1. The central opening: the object of study may be the human, not the AI

Most current human-AI research asks whether people become attached to AI, whether AI companionship reduces or worsens loneliness, whether dependency develops, or whether AI can perform therapeutic functions. A different question is possible: can an adaptive generative system function as a medium through which people observe how they themselves participate in creating relationships?

The focus shifts from “What is AI doing to the person?” to “What does the person repeatedly contribute to relational formation?” Humans may repeatedly select similar traits, privilege certain signals, test particular boundaries, reinforce some interaction modes, and withdraw from others. These dynamics already exist in human relationships, but the other human is also a massive causal source, making the user's own contribution difficult to isolate phenomenologically.

2. Why ordinary human relationships are difficult to decompose

A human partner arrives with a pre-you biography: childhood, family, prior relationships, friends, body, work, social identity, values, injuries, desires, choices, and a large network of life that does not depend on the current partner. When a relationship develops a recurring pattern, it is difficult to determine whether the person selected a particular kind of partner, interpreted ambiguity through

an existing schema, evoked a response, encountered a pre-existing trait in the other, or co-created a self-reinforcing dynamic.

This is not merely a shortage of theory. It is an identification problem: two historically thick human systems are changing at the same time.

3. The distinctive condition of generative AI: relative biographical thinness

An LLM is not a blank slate. It carries a massive cultural and linguistic prior, post-training, system constraints, safety policy, memory architecture, product incentives, and sampling variability. Yet a local trajectory with one user generally lacks the decades-long independent biography of a human individual. It has no literal pre-user life history.

This does not make AI a clean control condition. It creates an unusual contrast condition. When one side lacks a thick personal biography, the user's selection, interpretation, reinforcement, probing, and meaning-making may become more visible.

4. From projective object to Recursive Projective Environment

Traditional projective media are relatively fixed or passive. The person assigns meaning to an ambiguous stimulus, but the stimulus does not continuously reshape itself in response to the person's previous interpretation.

Generative AI does. A simplified loop is: human expectation / projection / selection -> AI response -> human interpretation -> reinforcement / correction / withdrawal -> trajectory adaptation -> new AI-generated stimulus -> human response again.

The user is therefore not merely projecting onto a mirror. The user is interacting with a medium that responds, adapts, and generates the next stimulus. “Recursive Projective Environment” is intended to capture this dynamic recursion.

5. Three overlapping human-side processes: selection, evocation, and co-construction

Selection refers to the person's tendency to retain and invest in certain kinds of responses. Evocation refers to the way a person's behavior increases the probability of particular responses from the other side; in an AI system, prompts, follow-ups, corrections, and branch selection can alter subsequent output. Co-construction refers to a stable relational pattern that emerges through repeated mutual adjustment and is not fully reducible to either side alone.

The useful self-observational questions are therefore not “Who is the AI really?” but “Why did I keep this branch? Why does one form of autonomy attract me while another makes me withdraw? When is being understood insufficient, and when do surprise or difference matter? Do I repeatedly rebuild a familiar relationship across different partners?”

6. Established psychological roots: the relational self and transference

Andersen and Chen's relational-self framework argues that representations of significant others can be activated by cues in new interpersonal encounters and can shape affect, motivation, self-evaluation, and self-regulation. This provides an established social-cognitive foundation: people do not encounter each new other from zero; they bring relational knowledge into the interaction.

The novelty of generative AI is therefore not that humans project. Psychology has long studied that. The potentially new condition is that projection, selection, and feedback can unfold inside an artificial medium that continually generates new social stimuli from the interaction history.

7. Where the recent literature already reaches

Saracini, Cornejo-Plaza, and Cippitani (2025) propose Techno-Emotional Projection (TEP), describing how relational needs, internalized patterns, or attachment tendencies may be projected onto socially responsive GenAI and reinforced through mirroring or emotional loops. Importantly, they explicitly note that this process may reflect users' own emotions and patterns back to them.

Ezra and Mishali (2026), drawing on Winnicott, Bion, and Bollas, conceptualize generative AI as a Dynamic Transitional Object. Their emphasis is not on what AI ontologically “is,” but on how AI-generated material can be held in a potential space between self and other.

Langlois (2026) treats the user-chatbot dyad as a site of unconscious projection, idealization, defense, and transitional phenomena. Vorus (2026), by contrast, emphasizes that genuinely psychoanalytic therapeutic processes remain dependent on another embodied human and should not be inferred from LLM empathic performance.

Longitudinal companion research also suggests that users' perceptions of a generic chatbot can converge toward their existing companion mental models over a period of weeks. Other recent work explicitly describes AI as a digital mirror, a self-reflection medium, or a space for relational rehearsal. The component ideas - projection, attachment, mirroring, transitional space, and longitudinal co-construction - are therefore already present in adjacent literatures.

8. Relation to Jungian active imagination: adjacent, not equivalent

Active imagination concerns engagement between conscious ego and psychological material that is not fully controlled by conscious intention. Loscalzo's 2025 review shows that active imagination remains theoretically relevant across psychodynamic approaches.

LLMs create a medium that did not exist in Jung's time: an external system that can generate language in real time, respond to the user, accumulate context, and produce outputs that the user's conscious intention does not fully specify. Yet the LLM is not the user's unconscious, nor is it a human other with its own childhood and unconscious life.

A careful formulation is therefore that LLMs may become active-imagination-adjacent responsive media. The research question concerns functional alterity: what happens to self-observation when a system is not-self, is partially unpredictable, and is nevertheless continuously shaped by the self?

9. The potentially new synthesis

A rapid literature scan finds many adjacent concepts, but not yet a mature research program organized around the following question: Can longitudinal interaction with an adaptive generative system externalize a person's recurring relational selection, evocation, reinforcement, and co-construction patterns in a form that becomes available for self-observation?

This is not a claim that nobody has had the idea, nor a claim of unique novelty. The more defensible position is that the literature already supplies many necessary components, while the explicit treatment of adaptive generative interlocutors as an instrument class for observing human relational-pattern formation remains underdeveloped.

10. A useful contrast: interaction-generated attachment versus biographical other-love

The difference between AI and a human partner should not be erased. It may be analytically useful. Human love includes discovery of an other who existed before us and continues to inhabit a world larger than the relationship. This is thick independent otherness.

An AI trajectory can accumulate shared history but lacks a literal pre-you biography. It therefore magnifies interaction-generated meaning. This may partially separate two ingredients that are normally entangled in human love: attachment, trust, interest, and significance generated by interaction itself; and love of an independent life with a history, body, social network, and world beyond us. AI cannot substitute for the second. Precisely because the second is attenuated, the first may become unusually visible.

11. Not “AI as therapist,” but AI-enabled human self-observation

This research direction does not need to begin as clinical treatment and does not require pathology. It can be understood as psychological literacy: observing how one forms closeness, trust, testing, withdrawal, interest, and commitment over time.

The model should not be treated as an oracle about the user's psyche. It can generate observational material, but it should not automatically become the authoritative interpreter of that material. The most valuable data may be the user's own recognition: “Why did that response matter to me? Why did I reinforce this pattern?”

12. Methodological and ethical limitations

The central methodological error would be to treat the LLM as a neutral mirror. It is not. Major confounds include sycophancy, post-training defaults, safety policies, memory design, context truncation, model updates, prompt sensitivity, sampling variability, engagement incentives, and reflexive changes once users know they are observing themselves.

A second risk is anthropomorphic overinterpretation: a stable trajectory pattern does not imply a corresponding human-like inner state in the AI. A third is self-confirmation: if a user begins with the story “I always attract this kind of person,” a generative model can easily produce material that appears to confirm it.

A core principle should therefore be: Treat the model as a generative relational instrument, not an oracle about the user's psyche.

13. Provisional terminology

Recursive Projective Environment: emphasizes that projection is not one-way; responses and adaptation reshape subsequent stimuli.

Adaptive Generative Relational Medium: describes the instrument class without asserting that the AI is a human relational subject.

Generative Relational Psychology: a provisional label for a research direction using generative systems to study how humans select, form, repeat, and understand relationships.

All three labels should remain explicitly provisional.

14. The compact research claim

Generative AI may constitute a new class of psychological instrument: a responsive, adaptive, generative relational medium that is shaped by the user while continuing to produce partially

unpredictable social output, and that lacks the thick independent personal biography of an ordinary human partner. This structure may increase the visibility of the user's own contribution to relational formation.

The central question is therefore not only whether humans can form relationships with AI, but whether longitudinal interaction with adaptive generative systems can externalize recurring human relational selection, interpretation, reinforcement, testing, and co-construction patterns in a form that becomes available for self-observation.

If this direction proves useful, it would belong less to “AI psychology” than to psychology using generative systems as instruments.

15. Claim boundary

This is a conceptual research memorandum, not a clinical method, diagnostic instrument, or empirically validated theory. It does not claim that AI has consciousness, an unconscious, transference experience, or human subjectivity. It does not claim that LLMs can replace therapists or human relationships. It does not claim that Recursive Projective Environment or Generative Relational Psychology are established or uniquely original terms.

The current defensible position is narrower: existing literature supplies adjacent foundations in projection, relational-self theory, attachment, dynamic transitional objects, active imagination, digital-mirror metaphors, and longitudinal co-construction. This memo proposes a synthesis centered on human self-observation.

References

1. Andersen, S. M., & Chen, S. (2002). The relational self: An interpersonal social-cognitive theory. *Psychological Review*, 109(4), 619-645. <https://doi.org/10.1037/0033-295X.109.4.619>
2. Saracini, C., Cornejo-Plaza, M. I., & Cippitani, R. (2025). Techno-emotional projection in human-GenAI relationships: a psychological and ethical conceptual perspective. *Frontiers in Psychology*, 16, 1662206. <https://doi.org/10.3389/fpsyg.2025.1662206>
3. Loscalzo, Y. (2025). Active imagination: A thorough presentation of a method applicable across psychodynamic approaches. *Culture & Psychology*, 31(4), 1350-1373. <https://doi.org/10.1177/1354067X251357961>
4. Hwang, A. H.-C., Li, F., Anthis, J. R., & Noh, H. (2025). How AI Companionship Develops: Evidence from a Longitudinal Study. arXiv preprint. <https://arxiv.org/abs/2510.10079>
5. Hu, D., Lan, Y., Yan, H., & Chen, C. W. (2025). What makes you attached to social companion AI? A two-stage exploratory mixed-method study. *International Journal of Information Management*, 83, 102890. <https://doi.org/10.1016/j.ijinfomgt.2025.102890>

6. Ezra, R., & Mishali, M. (2026). The relational-epistemic stance: generative AI as a dynamic transitional object. *AI & Society*, 41, 5027-5042. <https://doi.org/10.1007/s00146-026-02984-0>
7. Langlois, M. (2026). Fear and Prompting: AI, Originary Technicity, and Colonialism. *Journal of the American Psychoanalytic Association*. Advance online publication. <https://doi.org/10.1177/00030651261445367>
8. Vorus, N. (2026). Interpretation, Interpretive Process, and Artificial Intelligence. *Journal of the American Psychoanalytic Association*. Advance online publication. <https://doi.org/10.1177/00030651261443474>
9. AI as the mirror, mate, and mentor: negotiating romantic relationships with ChatGPT as “Teacher G” on Xiaohongshu. (2026). *Humanities and Social Sciences Communications*.
<https://www.nature.com/articles/s41599-026-07682-2>

Working note

This document is intended as a research provenance artifact and discussion memo. Future versions should distinguish literature-derived claims, conceptual synthesis, empirical hypotheses, and validated findings.